# 1 – Permutations & Combinations

*Tutor: Chin Yi*

# Introduction

- Chin Yi
- kokc0010@e.ntu.edu.sg

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MH1820 Tutorial

https://forms.cloud.microsoft/r/Y5jgUkwqmB

# Recap: Ordered Sampling

**Theorem 5**

*The total number of ordered sample of size $k$ from a set of $n$ different objects, with replacement, is*

$$n^k.$$

**Theorem 8 (Number of permutation)**

*A set with $n$ different objects has exactly $n!$ permutations.*

*Here, $n! = n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1$ ($n$-factorial).*

**Theorem 9 (Number of $k$-permutations)**

*A set with $n$ different objects has exactly*

$$P(n,k) = \frac{n!}{(n-k)!}$$

*$k$-permutations.*

# Recap: Unordered Sampling

**Theorem 11 (Number of combinations)**

*The number of $k$-element combinations of a set with $n$ different objects is*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

$$\binom{n}{k} = \binom{n}{n-k}$$

# Recap: Probability of Event

**Theorem 19 (Finite sample space with fair outcomes)**

*Let $\Omega$ be a finite sample space, and $A$ be an event. Assume that all possible outcomes have the same probability, The probability of the event $A$ is given by*

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

**Problem 1 ($n$-tuples, Multiplication principle)**

(a) How many 7-digit numbers are there all of whose digits are odd?

(b) How many 5-digit numbers are there all of whose digits are primes? (a prime is an integer $p \geq 2$ whose only divisors are 1 and $p$; note that 0 and 1 are not primes)

(c) How many 5-digit numbers are there? (all digits are allowed, but the first digit must not be 0)

(d) If a 5-digit number is chosen randomly, what is the probability that all its digits are primes?

(e) How many 6-digit numbers are there with no repeated digits?

(f) If a 6-digit number is chosen randomly, what is the probability that it has no repeated digits?

# Bonus Qn: License plates

(1) (a) How many different 7-place license plates are possible if the first 2 places are for letters and the other 5 for numbers?

(b) Repeat part (a) under the assumption that no letter or number can be repeated in a single license plate.

**Problem 2 (Permutations)**

(a) How many possibilities are there to line up 5 persons in a queue?

(b) Suppose there are 2000 spectators in a soccer-stadium. During halftime, a queue of 5 spectators forms in front of a coffee shop. How many possibilities are there for such a queue to form from the spectators?

# Relationship between Combination and Permutations

**Theorem 11 (Number of combinations)**

*The number of $k$-element combinations of a set with $n$ different objects is*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

**Theorem 9 (Number of $k$-permutations)**

*A set with $n$ different objects has exactly*

$$P(n, k) = \frac{n!}{(n-k)!}$$

*$k$-permutations.*

P(n, k) = C(n, k) * k!

## Problem 3 (Combinations)

We consider a standard poker deck with 52 cards:

(a) How many ways are there to select a poker hand of 5 cards from the deck of 52 cards?

(b) What is the probability that a randomly chosen poker hand of 5 cards forms a
(i) flush (excluding straight flush) (ii) pair, (iii) two pairs, (iv) full house?

(c) What is the probability that a randomly chosen hand forms a full house without any aces or kings?

Bonus Qn: How many ways are there to select a poker hand of 5 cards from the deck of 52 cards where they have the same colour (ie. All 5 cards are either BLACK or RED).

**Problem 4 (Permutations of multisets)**

(a) How many words can be formed from the letters in `successlessness`? (here every permutation of these letters counts as a "word" even if does not make sense)

(b) In an orchid show, 11 orchids are to be placed along one side of the greenhouse. There are 5 lavender orchids, 4 white orchids, and 2 yellow orchids. Considering only the color of the orchids, find the number of different color displays?

(c) How many possibilities are there to split up 12 players into Team $A$, $B$, $C$ with 5, 4, 3 players respectively?